

EMPIRICAL EVALUATION OF THE FOLLOW-THE-REGULARIZED-LEADER ALGORITHM FOR THE PURPOSE OF ONLINE PORTFOLIO OPTIMIZATION

Prerana Kolipaka

School of Electrical and Computer Engineering, Georgia Institute of Technology

ABSTRACT:

Online Portfolio Optimization involves strategically choosing the most efficient mix of diverse financial assets in real-time. This project focuses on implementing the Follow-The-Regularized-Leader (FTRL) algorithm to execute this optimization task. FTRL is an online machine learning optimization technique used in the realm of online advertising, recommendation systems, etc. This implementation of the algorithm involves applying this algorithm to a historical share price dataset that contains the data of top 100 companies listed in the Australian Securities Exchange. The FTRL algorithm showcases its prowess by minimizing regrets, effectively reducing incurred losses. A comparative analysis is conducted between this algorithm and the FTL algorithm to evaluate their performance.

INTRODUCTION:

Portfolio optimization is the process of using mathematical approach to select the optimal combination of investment portfolio assets and their values, given a set of objectives and constraints. The goal of Online Portfolio Optimization (OPO) is to reduce the loss (minimize risk), in turn maximizing the returns in real-time as new data becomes available. The task is carried out via a variety of online optimization techniques, including FTL, FTRL, FTRL-p, Online Gradient Descent, Kalman Filters, etc. Follow-The-Leader (FTL) algorithm involves learning from decisions made by the leader in the past. It relies on exploiting past data; there's no exploration. However, in adversarial scenarios like Online Portfolio Optimization, relying only on past data might not give the best results. There needs to be some randomization. FTRL is an algorithm with an added regularization term that

balances the exploitation and exploration factor of the online algorithm resulting in optimized decisions for adversarial scenarios like the Portfolio Optimization.

METHODOLOGY:

Follow-The-Leader Algorithm:

FTL algorithm learns from the past data and follows the "leader" that made these decisions. The weights in FTL algorithm are updated according to the following equation:

$$\omega_t = \arg \min_{\omega \in B} [L_{t-1}(\omega)] \quad (1)$$

Follow-The-Regularized-Leader Algorithm:

FTRL algorithm is a Follow-The-Leader (FTL) Algorithm with a regularization term added. FTL algorithm is an exploitative algorithm that purely uses the past data to make a decision. Adding a regularization term to the FTL algorithm helps achieve this task of decision making in adversarial applications.

Modelling the problem:

In the application of Online Portfolio Optimization, the problem is modelled as follows:

Consider a scenario where we are playing the market for T days managing a portfolio consisting of i stocks. Every day we decide to buy or short certain number of shares of each stock. We make a decision to buy or short a number of stocks every morning. At the end of the day, we sell whatever we bought and buy back the stock that we shorted.

Let, $\mathbf{w}_t$ be a 1-D vector of size i , containing the decision. Here t indexes the day and every

element represent the number of shares bought or sold of each stock. D represents the bound on the number of stocks of each type that can be bought or shorted. $\mathbf{l}_t$ represents the loss vector, that represents how much the stock depreciated. It is calculated as (opening price – closing price).

Now, the FTRL weight update is performed according to the equation:

$$\omega_t = \arg \min_{\omega \in B} [L_{t-1}(\omega) + 1/\eta \|\omega\|_2^2] \quad (2),$$

where, $L_{t-1}(\omega) = \sum_{s=1}^{t-1} l_s(\omega)$ and B is the ∞ ball of D .

The performance of the algorithm is evaluated by its Regret. Regret is the measure of the algorithm's performance in comparison with the to an ideal scenario or the best fixed decision in hindsight.

The Regret for FTRL is calculated as:

$$R_T := H_T - L_T^* \quad (3),$$

where,

$$L_T^* := \min_{\omega \in B} \sum_{t=1}^T \langle \omega, l_t \rangle \quad (4),$$

and

$$H_T := \sum_{t=1}^T \langle \omega_t, l_t \rangle \quad (5)$$

L_T^* represents the loss obtained by using the best fixed decision in hindsight. H_T represents the total loss incurred by the FTRL algorithm.

The value of η chosen for this application is dynamic, that is $\eta = \frac{D\sqrt{i}}{G\sqrt{T}}$ that achieves a low regret of order $O(\sqrt{i}DG\sqrt{T})$. Here G is the upper bound on losses: $\|\mathbf{l}_t\|_2 \leq G$.

Regret Bound Calculations:

As mentioned in the lecture notes, the Regret term is divided into parts as:

$$R_T = \underbrace{H_T - L_T(\tilde{\omega}^*)}_{R_T^A} + \underbrace{L_T(\tilde{\omega}^*) - L_T^*}_{R_T^B}.$$

R_T^B is bounded by:

$$R_T^B \leq D^2 / \eta$$

In this implementation, there is a limit on number of each stock that can be bought as opposed to the limit D on overall buying capacity as mentioned in the notes. Therefore, the R_T^A bound changes to:

$$R_T^B \leq (iD^2) / \eta \quad (6)$$

Similarly, the R_T^A bound changes to:

$$R_T^A \leq (iD^2) / \eta + \eta G^2(T+1) \quad (7)$$

So, the Regret bound is:

$$R_T \leq (2*iD^2) / \eta + \eta G^2(T+1) \quad (8)$$

$$\leq O(\sqrt{i}DG\sqrt{T}). \quad (9)$$

This Regret bound is obtained by setting $\eta = \frac{D\sqrt{i}}{G\sqrt{T}}$.

DATASETS:

The Australian Historical Stock Prices dataset is used to implement this algorithm. The dataset contains historical share price data from top 100 companies listed on the Australian Securities Exchange. The dataset contains information including date, opening price, closing price, high price, low price, adjusted closing price and trading volume. For this task, we only use the date, opening price and closing price information.

IMPLEMENTATION:

For the implementation, the value of D is set to 10. Since the value of shares are less than 10 AUD, $D=10$ is a reasonable limit. The implementation is done for $i=5$ stocks (first result graph) and $i=2$ stocks (for rest of the graphs). For the first result, the algorithm is iterated over 300 days, while for the rest of the graphs the algorithm is iterated over 500 days. The value of

η chosen for this application is dynamic, that is $\eta = \frac{D\sqrt{i}}{G\sqrt{T}}$.

RESULTS:

Results obtained by simulating the algorithm:

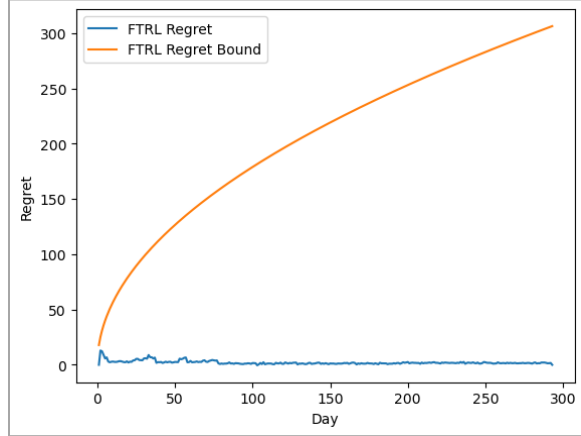


Fig 1: FTRL regret over time; $i=5$, $D=10$, $T=300$

From this graph, we can see that the FTRL regret is well below the Regret bound. This could be because the stocks do not vary too much.

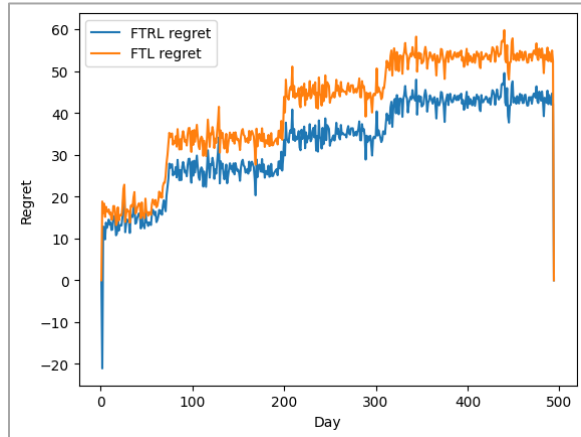


Fig 2: Comparison of the Regret of FTRL and FTL algorithm; $i=2$, $D=10$, $T=500$

From this graph, it can be concluded that FTRL achieves lower regret, hence performs better than the FTL algorithm.

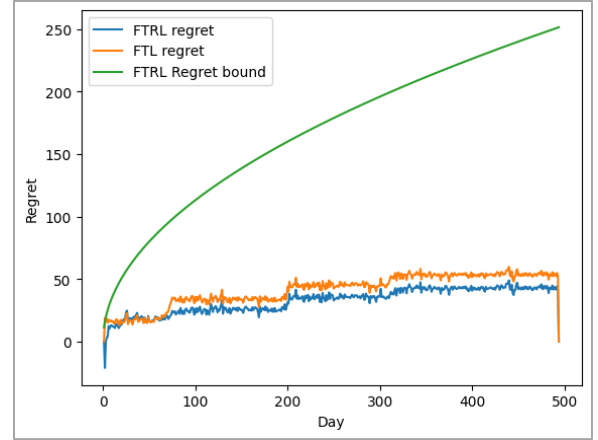


Fig 3: Comparison of FTRL and FTL graph with the FTRL regret bound; $i=2$, $D=10$, $T=500$

We see that both the algorithms achieve low regret as compared to the bound.

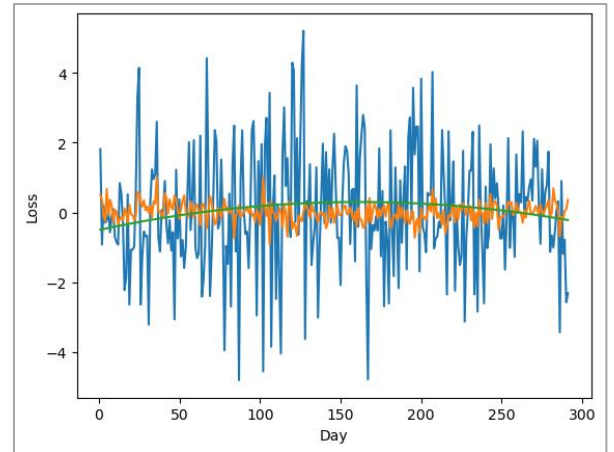


Fig 4: FTRL loss graph

This graph denotes the total loss occurred on each day. We can see that the variation is between 4 and -4 only. This shows that the losses do not spike at any point.

CONCLUSIONS:

Based on the results, it is evident that the FTRL algorithm surpasses the FTL algorithm in performance. The FTRL algorithm demonstrates notably lower regret compared to its FTL algorithm. The implementation adheres closely to the regret bound of the FTRL algorithm, noted as $O(\sqrt{i}DG\sqrt{T})$. The algorithm effectively

minimizes loss, as observed by the limited increase in loss, which does not exceed 4 points. However, FTRL algorithm becomes computationally expensive as the number of financial instruments increases.

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- Lecture notes (Lecture 3,4,8 and 9)
- Dataset: <https://www.kaggle.com/datasets/ashbellett/australian-historical-stock-prices>